Project Report Format

1. **INTRODUCTION**

1.1 Project Overview

The project **“Measuring the Pulse of Prosperity: An Index of Economic Freedom Analysis”** focuses on analyzing and visualizing global economic freedom using data-driven techniques. Economic freedom is a critical factor in understanding a country’s prosperity, stability, and development. The project leverages tools such as **Python**, **MySQL**, and **Tableau** to clean, process, and visualize data derived from the **Index of Economic Freedom**, published annually by The Heritage Foundation.

By breaking down the index into its four main pillars—**Rule of Law, Government Size, Regulatory Efficiency, and Open Markets**—this project enables a comprehensive exploration of global economic conditions.

1.2 Purpose

- To **simplify and visualize** complex economic data for researchers, analysts, and policymakers.
- To help users **explore correlations** between economic indicators like inflation, GDP, unemployment, and economic freedom.
- To build an **interactive dashboard** that allows filtering and comparing countries across various metrics and years.
- To provide **insights** that support data-driven decisions in policy-making, investment, and academic research.
- To foster **economic awareness and education** through a user-friendly, data-centric approach.

2. **IDEATION PHASE**

2.1 Problem Statement

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

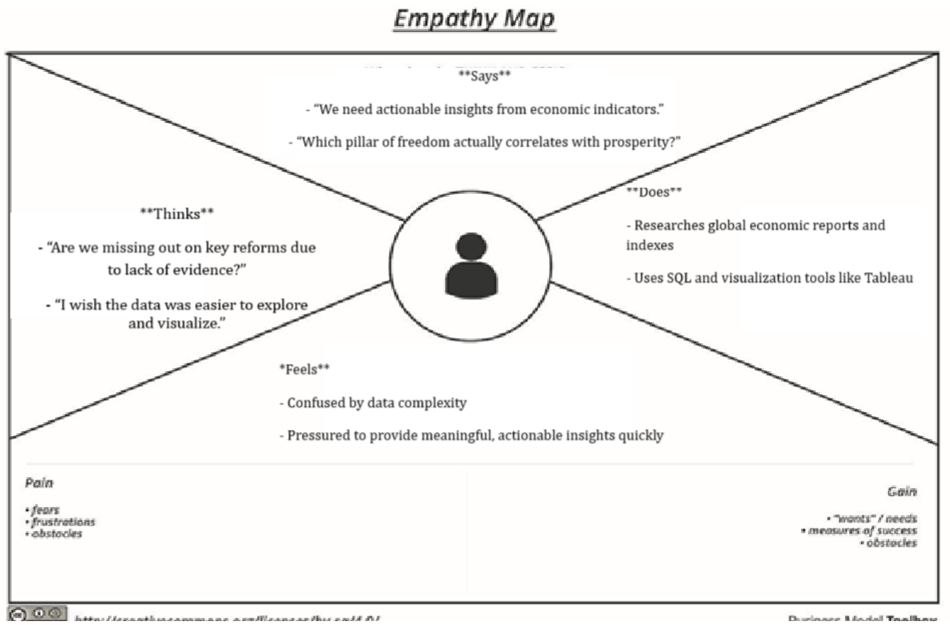
A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you’ll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - *who are they?*	Describe the customer and their attributes here
I'm trying to	*List their outcome or "job" the care about - what are they trying to achieve?*	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – *what bothers them most?*	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?*	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – *how does it impact them emotionally?*	Describe the emotions the result from experiencing the problems or barriers

Reference: https://miro.com/templates/customer-problem-statement/

Problem Statement (PS)	**I am (Customer)**	**I'm trying to**	**But**	**Because**	**Which makes me feel**
PS-1	I am a global policy researcher or economic advisor	I'm trying to explore how economic freedom influences national prosperity and investment appeal	But I lack integrated, visual insights to analyze the multidimensional data effectively	Because existing data is often fragmented, complex, and not easy to interpret in raw formats	Which makes me feel uncertain about prioritizing the right policy reforms
S-2	I am a data analyst or economist in a development agency	I'm trying to create dashboards and reports showing the economic status of different countries	But I can't easily compare indicators like GDP, HDI, and economic freedom scores in one view	Because there's no unified platform to visualize and analyze cross-country economic freedom data	Which makes me feel frustrated and restricted in presenting meaningful insights

2.2 Empathy Map

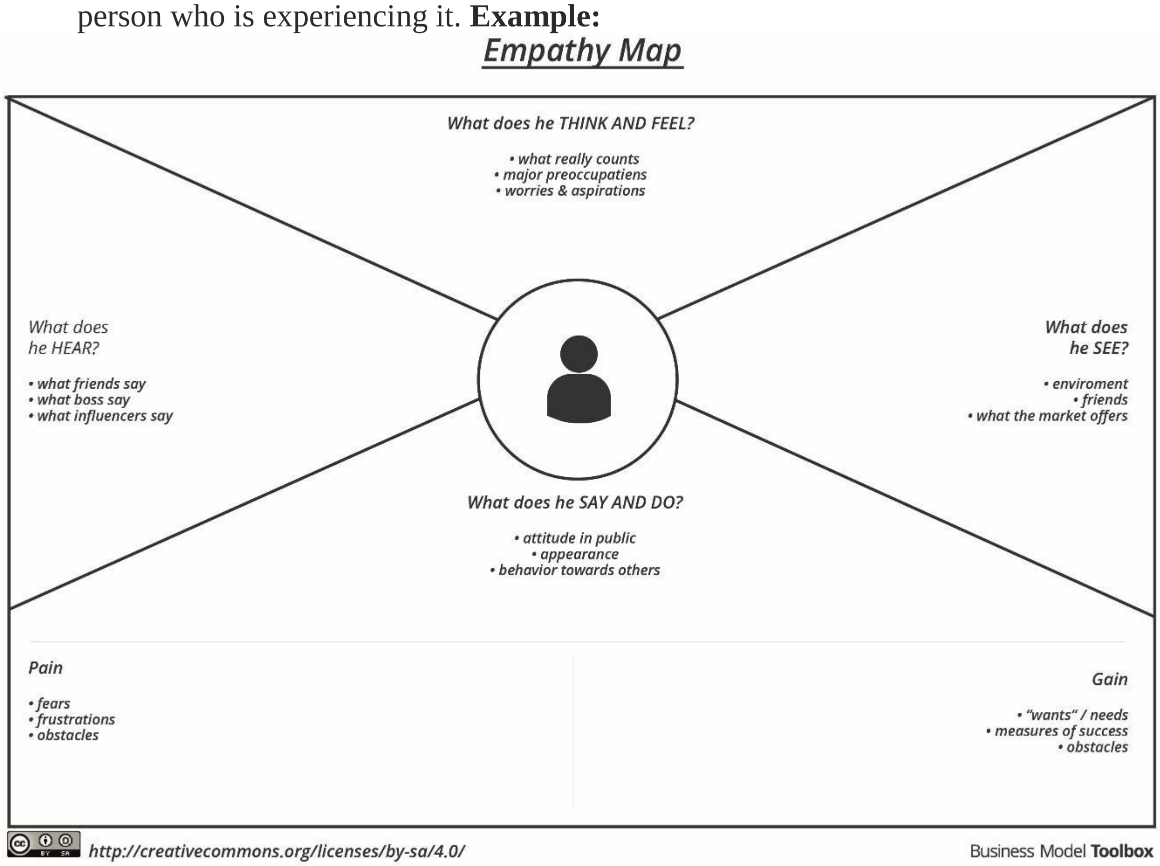


2.2Empathy Map Canvas

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. **Example:**



Reference: https://www.mural.co/templates/empathy-map-canvas

2.3Brainstorming

Brainstorm & Idea Prioritization Template

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Problem Statement:

How can data analytics be used to explore and visualize the relationship between economic freedom and national prosperity, using the Index of Economic Freedom and complementary global economic indicators?

- Goals of the Project:

- Perform exploratory data analysis using MySQL.
- Integrate and manage data from multiple economic sources including the Index of Economic Freedom, World Bank, and IMF.
- Visualize economic freedom indices and prosperity metrics using Tableau.
- Identify patterns and draw actionable insights about how economic freedom correlates with prosperity.

Step-2: Brainstorm, Idea Listing and Grouping

- Data Collection & Preparation

- Import Index of Economic Freedom dataset into MySQL.
- Clean and normalize data (handle nulls, consistent formatting).
- Categorize data into the 4 pillars: Rule of Law, Government Size, Regulatory Efficiency, Open Markets.
- Integrate complementary datasets (GDP, HDI, Poverty Rates).

- Database Design & Querying

- Design a normalized database schema to store country-level indicators.
- Write SQL queries to find top/bottom countries by overall score.
- Create views for regional and income-level comparisons.

- Visualization & Dashboard Creation (Tableau)

- Develop a world heat map of economic freedom scores.
- Create bar charts comparing pillars across top 10 and bottom 10 countries.
- Design correlation scatter plots between economic freedom and GDP/HDI.
- Build interactive time-series visualizations if historical data is available.

- Insight Generation & Analysis

  - Analyze regional trends in economic freedom.
  - Determine which pillars are most strongly associated with prosperity.
  - Identify outliers where high freedom does not equate to high prosperity (or vice versa).
  - Draft recommendations based on findings.

- Advanced (Optional)

  - Run regression or clustering models to predict economic performance.
  - Forecast future trends based on current data

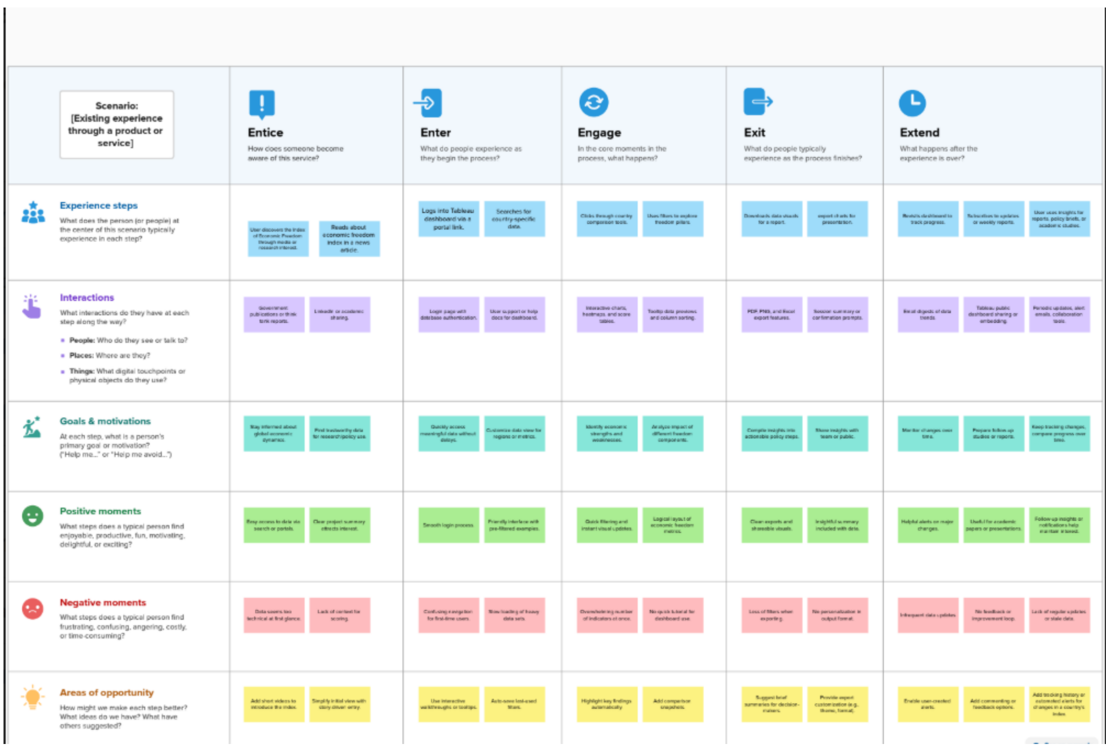
Step-3: Idea Prioritization

Idea/Task	Impact	Feasibility	Priority
Cleaning and importing data into MySQL	High	High	High
Creating structured schema for 4 economic freedom pillars	High	High	High
Writing SQL queries for rankings and comparisons	Medium	High	High
Adding GDP, HDI data from World Bank for deeper insights	High	Medium	High
Creating a Tableau dashboard with	High	Medium	High

map and correlation plots			
Running regression or clustering (advanced analysis)	Medium	Low	Medium
Comparing countries over time (if multiple years present)	High	Medium	Medium
Analyzing outliers and exceptions in data	Medium	Medium	Medium

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIN
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Data Upload & Integration	Upload CSV/XLSX datasets Connect to MySQL Database Validate column headers and formats
FR-4	Data Visualization	Build interactive dashboards using Tableau Support filtering by country, year, and indicator Enable hover tooltips and drill-down charts

FR-5	Data Analysis & Reporting	Perform Exploratory Data Analysis (EDA) Generate insights and trends by region and indicator Export findings to PDF/CSV
FR-6	User Interaction	Save dashboard views Share visualizations via link or export Provide search and comparison features
FR-7	Notification & Updates	Send alerts on new data releases Notify users about changes in economic indicators

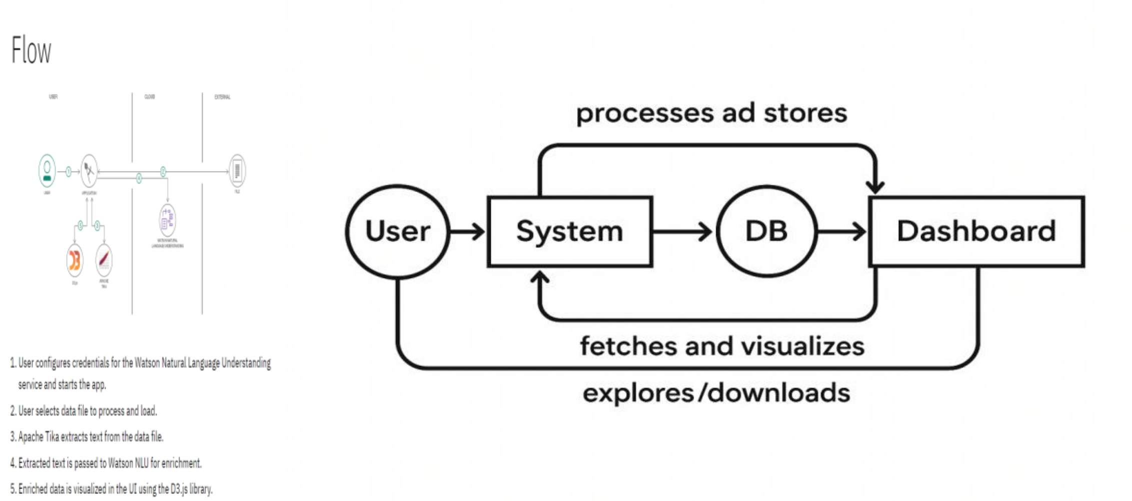
Non-functional Requirements:
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	User-friendly dashboard with intuitive filters and search features.
NFR-2	**Security**	Secure login and data access using HTTPS, Gmail/LinkedIn authentication.
NFR-3	**Reliability**	Ensure consistent performance during analysis and exports.
NFR-4	**Performance**	Fast data loading and real-time dashboard updates.
NFR-5	**Availability**	Fast data loading and real-time dashboard updates.
NFR-6	**Scalability**	Support increasing user and data volume with minimal lag.

3.3Data Flow Diagram

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example **(Simplified)**



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Analyst (Web User)	Registration	USN-1	As a user, I can register using email and password.	Able to access dashboard after login.	High	Sprint-1
Analyst (Web User)	Registration	USN-2	As a user, I receive confirmation via email after registering.	Confirmation email is sent and link works.	High	Sprint-1
Analyst	Login	USN-3	As a user, I can log in with registered credentials.	Able to log in securely and access dashboard.	High	Sprint-1
Analyst	Data Upload	USN-4	As a user, I can upload economic freedom data in CSV format.	System accepts and stores valid data.	High	Sprint-2
Analyst	Data Integration	USN-5	As a user, I can connect to a MySQL database.	Connection is established successfully.	Medium	Sprint-2
Analyst	Data Visualization	USN-6	As a user, I can view dynamic dashboards using Tableau.	Dashboards respond to filters and user actions.	High	Sprint-2
Analyst	Filtering & Search	USN-7	As a user, I can filter data by year, region, and indicator.	Results adjust in real-time based on filters.	Medium	Sprint-2
Analyst	Export	USN-8	As a user, I can download charts and reports from the dashboard.	Reports are downloadable in PDF/CSV formats.	Medium	Sprint-3
Admin	Management	USN-9	As an admin, I can manage user accounts and uploaded data.	Admin dashboard lists and controls access.	Medium	Sprint-3

3.4Technology Stack

Technical Architecture:

The system collects structured datasets (e.g. CSV/Excel) on economic freedom, stores them in a relational database (MySQL), and presents them through a web-based interactive dashboard built with Tableau. Data is processed and cleaned using Python for visualization purposes.
Infrastructure: Hybrid (Local for development, Cloud for visualization hosting if applicable)
External Interfaces: Economic data sources (Heritage Index, IMF APIs if used)

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web-based interactive dashboards	Tableau
2	Application Logic-1	Data pre-processing & EDA logic	Python (Pandas, NumPy, Matplotlib)
3	Application Logic-2	Data transformation & connection to Tableau	Tableau Prep / Python

4	Database	Relational database for storing economic data	MySQL
5	Cloud Database	Optionally used if deployed in cloud	Amazon RDS / Google Cloud SQL
6	File Storage	CSV/Excel files of economic data	Local Storage / Cloud Drive
7	External API-1	Used for fetching global economic data (optional)	IMF API / Heritage API
8	Infrastructure	Deployment environment	Local (Dev), Cloud (Deployment via Tableau Public / Server)

Table-2: Application Characteristics:

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Python packages used for data cleaning & visualization	Pandas, NumPy, Seaborn
2	Security Implementations	Securing access to dashboards & data exports	Tableau Access Permissions, Authentication via Email
3	Scalable Architecture	Dashboards & database can scale horizontally	Tableau Server / Tableau Cloud, MySQL Scaling
4	Availability	Hosted dashboards are accessible 24/7 online	Tableau Public / Cloud-hosted reports
5	Performance	Quick dashboard load, filter optimization, efficient querying	Tableau Extracts, Indexed MySQL Tables

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:


Problem-Solution fit canvas 2.0 Purpose / Vision

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS
Who is your customer?

Policy makers and government think tanks

Economics students and educational institutions

Researchers, data analysts, economists

6. CUSTOMER CC
What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

Lack of technical skills in data analysis

Limited access to cleaned or organized datasets

Inconsistent internet availability in rural institutions

5. AVAILABLE SOLUTIONS AS
Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

Static PDFs or Excel reports by Heritage Foundation, IMF, World Bank

Country-wise rankings with limited interactivity

Generic dashboards with minimal customization options

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS J&P
Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

Compare and analyze economic freedom across countries

Understand factors that influence prosperity

Present economic insights in an engaging, visual format

9. PROBLEM ROOT CAUSE RC
What is the real reason that this problem exists? What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

Dispersed, unstructured economic data

Lack of accessible tools for visualizing multi-country comparisons

Need for real-time, flexible analysis that static tools cannot fulfill

7. BEHAVIOUR BE
What does your customer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

Users download raw data, manually clean/analyze in Excel

Refer to yearly economic reports for country comparisons

Attend economic briefings or webinars to extract trends

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS TR
What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.

ınterest in global economic rankings and development trends

4. EMOTIONS: BEFORE / AFTER EM
How do customers feel when they face a problem or a job and afterwards?
i.e. lost, insecure > confident, in control - use it in your communication strategy & design.

Before: Confused, overwhelmed by data, disconnected from trends

After: Confident, clear insights, visually informed decisions

10. YOUR SOLUTION SL
If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

Filtering by country, pillar, and year

Real-time comparison and insights

Exporting of charts for policymaking or reports

8. CHANNELS of BEHAVIOUR CH
8.1 ONLINE
What kind of actions do customers take online? Extract online channels from #7

Online: Use Tableau dashboards, educational platforms, research portals

8.2 OFFLINE
What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

Offline: Refer printed economic reports, classroom discussions, policy briefings

Extract online & offline CH of BE

References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe

4.2 Proposed Solution

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Lack of interactive, centralized tools to analyze economic freedom data, making it difficult for researchers and policymakers to extract meaningful insights from large and scattered datasets.
2.	Idea / Solution description	Build a dynamic and user-friendly Tableau dashboard backed by MySQL and powered by Python scripts. The system enables real-time filtering, cross-country comparisons, and data exports across 12 indicators of economic freedom.
3.	Novelty / Uniqueness	Combines data visualization with real-time analytics on global economic freedom—making static, complex datasets accessible, interactive, and insightful for decision-makers.
4.	Social Impact / Customer Satisfaction	Empowers researchers, students, and policy advisors with accessible economic tools that support informed decisions and promote transparency in governance and education.

5.	Business Model (Revenue Model)	Preemium model: Basic version is open-access; premium features like country-specific reports, PDF exports, and advanced insights available via subscription for academic or policy institutions.
6.	Scalability of the Solution	Highly scalable across regions and years—can incorporate more datasets like GDP, HDI, or inflation and expand visualization capacity using Tableau Server or cloud deployment.

4.3Solution Architecture

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

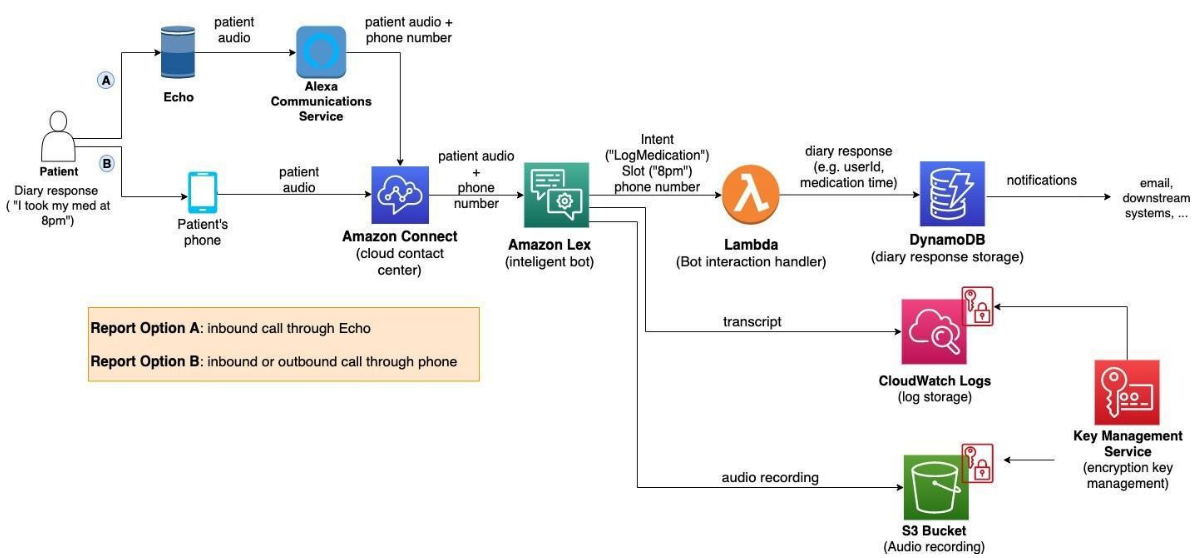
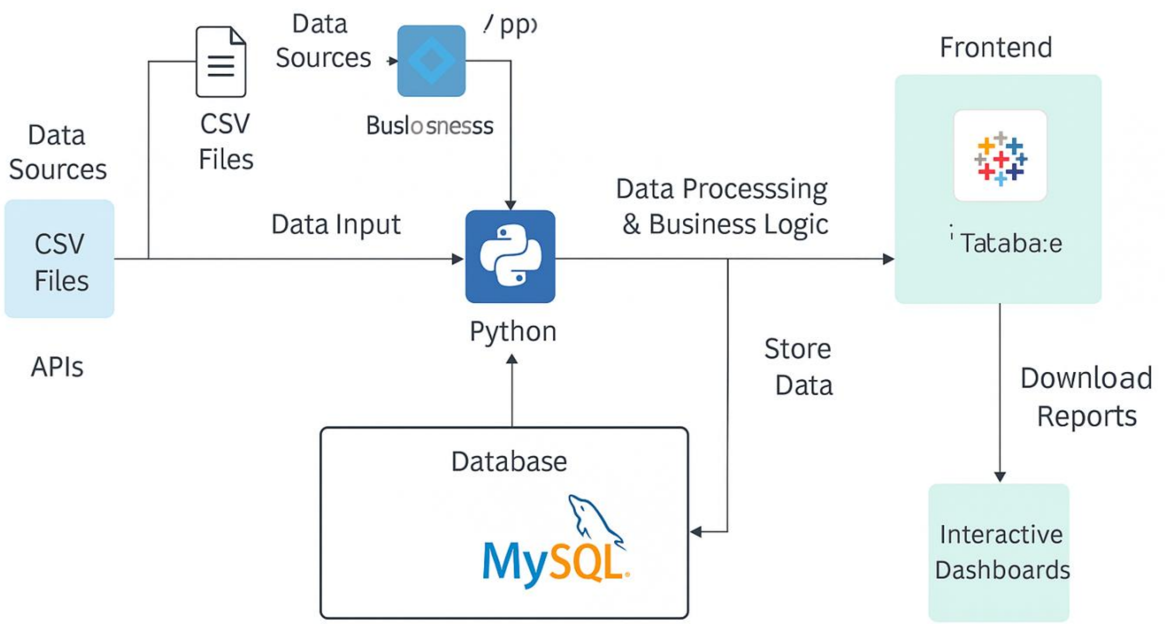


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/

Solution Architecture Diagram: Measuring the Pulse of Prosperity: An Index of Economic Freedom Analysis



5. PROJECT PLANNING & SCHEDULING

5.1Project Planning

Product Backlog, Sprint Schedule, and Estimation (4 Marks)
Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	2	High
Sprint-1	Registration	USN-2	As a user, I will receive confirmation email once I have registered for the application	1	High
Sprint-2	Registration	USN-3	As a user, I can register for the application through Facebook	3	Low
Sprint-1	Registration	USN-4	As a user, I can register for the application through Gmail	2	Medium
Sprint-1	Login	USN-5	As a user, I can log into the application by entering email & password	1	High
Sprint-3	Visualization	USN-6	As a user, I want to create dashboards in Tableau showing performance across the 4 pillars of economic freedom.	6	High

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-4	Reporting & Insights	USN-7	As a user, I want to generate reports summarizing findings with visualizations and insights.	8	High
Sprint-2	Data Analysis & Insights	USN-3	As a user, I want to analyze economic freedom indicators (12 factors) by region and income	3	Medium

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	6	6 Days	14-6-2025	19-6-2025	6	20-6-2025
Sprint-2	6	6 Days	20-6-2025	25-6-2025	6	25-6-2025
Sprint-3	6	6 Days	26-6-2025	1-7-2025	6	28-6-2025
Sprint-4	8	3 Days	1-7-2025	3-7-2025	8	28-6-2025

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

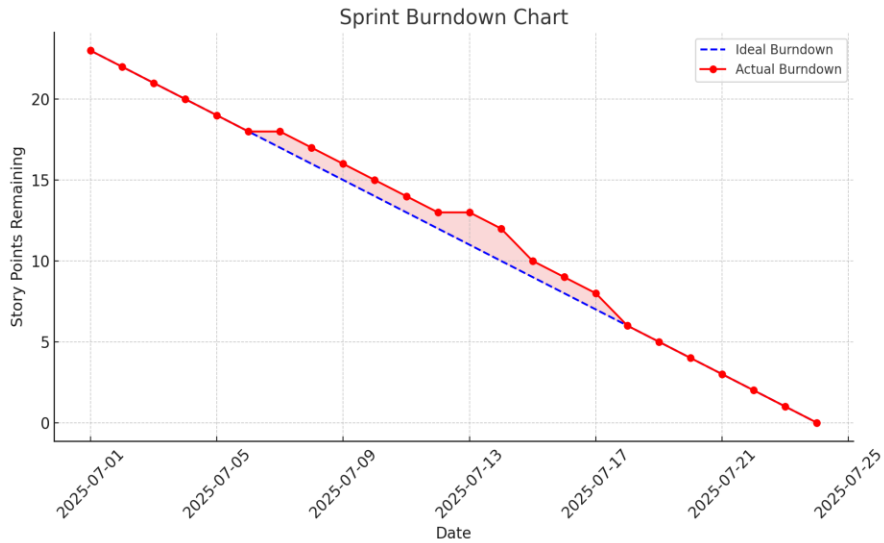
Sprint Duration = 6 days and average story points = 6

Then,

Velocity = 6 story points / 6 days = 1 point/day

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.



https://www.visual-paradigm.com/scrum/scrum-burndown-chart/

https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/project-management/estimation

https://www.atlassian.com/agile/tutorials/burndown-charts

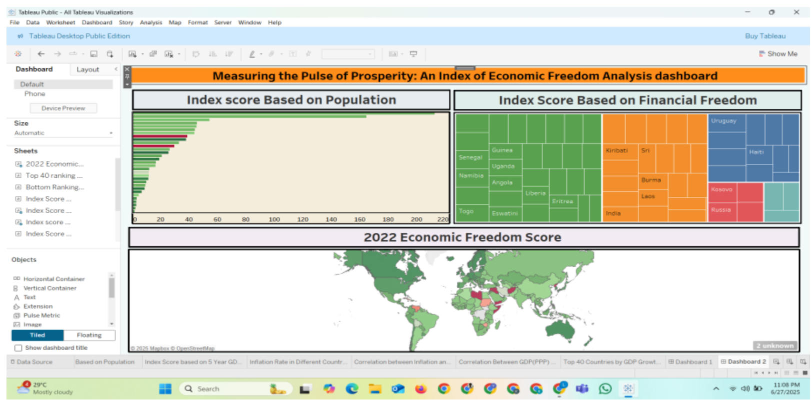
6. FUNCTIONAL AND PERFORMANCE TESTING

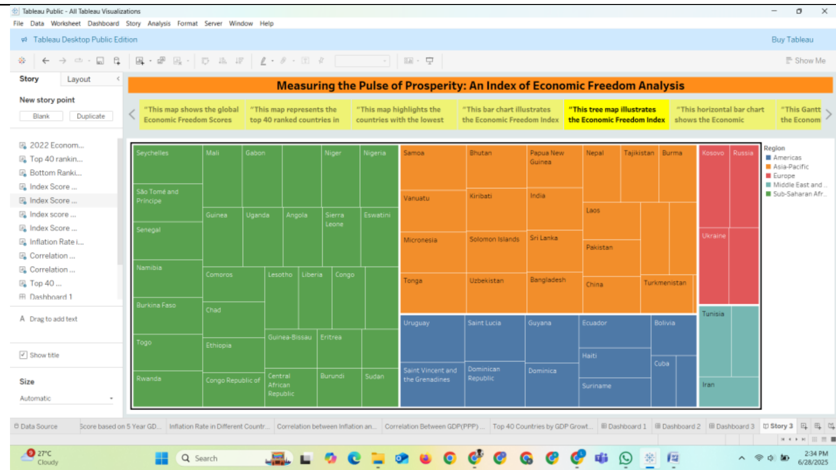
6.1 Performance Testing

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	The dataset includes 180+ countries with scores across 12 indicators and additional metadata like region, GDP, and population.

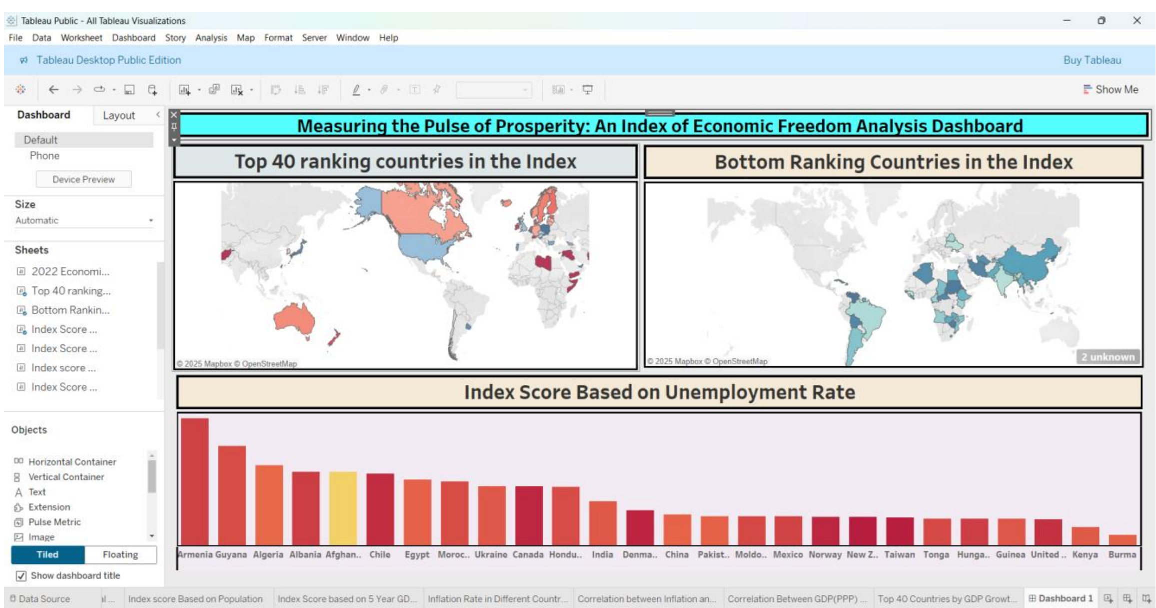
2.	Data Preprocessing	Data cleaning (null values, incorrect types) Normalization of scores Derived regional classifications
3.	Utilization of Filters	Region filter (Asia, Europe, Americas, etc.) Score filter (range slider for economic freedom score) Income level
4.	Calculation fields Used	**Average Score**: Across all 12 indicators **Freedom Category**: Derived as Low, Moderate, High based on total score **Performance Rank**: Dynamic country ranking by selected indicator
5.	Dashboard design	No of Visualizations / Graphs – 11 **2022 Economic freedom score** **Top 40 ranking countries in the index** **Bottom ranking countries in the index** **Index score based on unemployment rate** **Index score based on financial freedom** **Index score based on population** **Index score based on 5 year on GDP growth rate(%)**

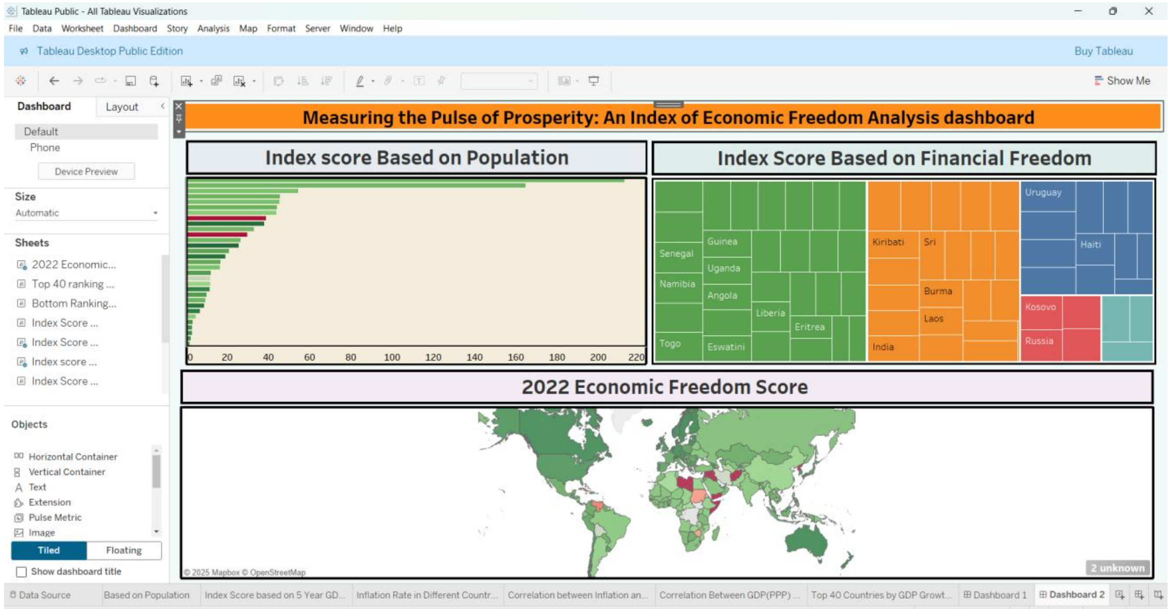
		 
6	Story Design	No of Visualizations / Graphs –11 

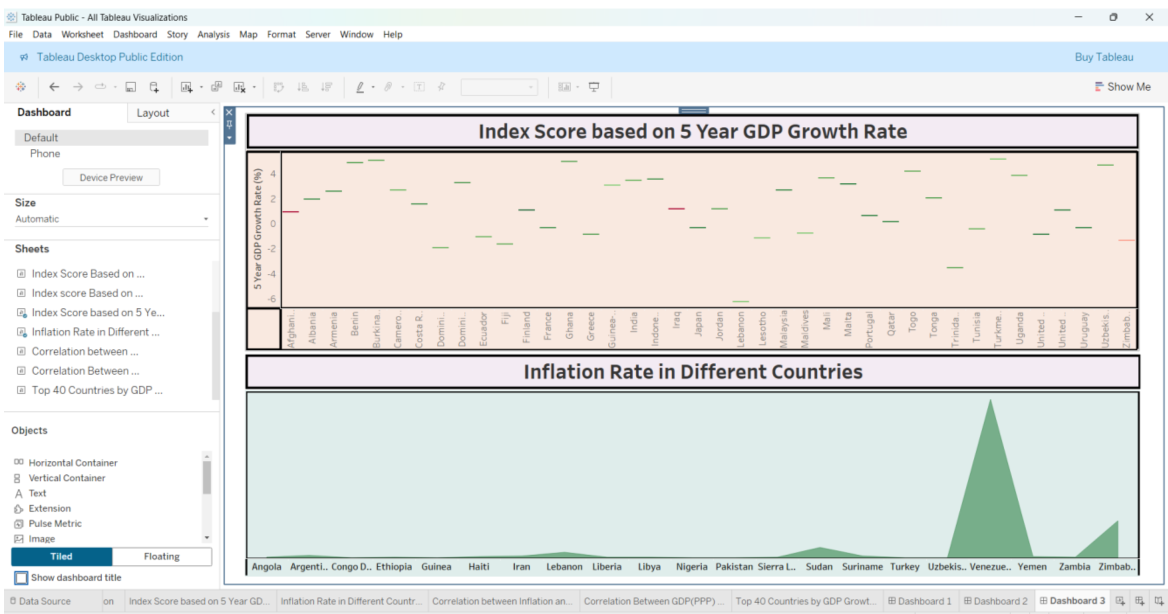
		

7. RESULTS

7.1 Output Screenshots







8. ADVANTAGES & DISADVANTAGES

Advantages

1. **Interactive Visualization**
   Enables users to explore complex economic data easily using dynamic dashboards in Tableau.
2. **Data-Driven Insights**
   Supports informed decision-making for policymakers, economists, and researchers.
3. **User-Friendly Interface**
   The dashboards are intuitive, requiring no advanced technical skills to use.
4. **Custom Filtering and Comparison**
   Allows comparison across countries, years, and economic indicators.

5. **Scalable and Expandable**
   Can be scaled to include additional indicators like GDP, HDI, or inflation trends.
6. **Educational Value**
   Helps students and scholars understand real-world economic concepts through visual learning.

Disadvantages

1. **Data Dependency**
   Accuracy depends on the availability and quality of external data sources like Heritage Foundation or IMF.
2. **Limited Offline Access**
   Tableau dashboards may require internet access for real-time interaction.
3. **Initial Setup Complexity**
   Requires knowledge of Python, MySQL, and Tableau for development and maintenance.
4. **Static for Non-Interactive Users**
   Users accessing exported charts or PDFs won't benefit from interactive features.
5. **Data Interpretation Required**
   Users must still understand basic economic terms and indicators to derive meaningful conclusions.

9. CONCLUSION

The project successfully leverages data analytics and visualization to provide meaningful insights into the global state of economic freedom. By utilizing tools like **Python, MySQL, and Tableau**, we transformed complex datasets into interactive dashboards that are easy to explore and understand.

Through this analysis, users can compare countries, detect economic trends, and assess the influence of factors such as inflation, unemployment, and GDP on a nation's economic freedom. The project not only empowers policymakers and researchers with actionable intelligence but also enhances public awareness about how economic policies affect prosperity and development.

Overall, this solution bridges the gap between raw economic data and informed decision-making, promoting transparency, education, and smarter economic governance.

10.FUTURE SCOPE

1. **Integration with Real-Time Data Sources**
   - Connect the system to APIs from organizations like the IMF, World Bank, or Heritage Foundation for automatic data updates.
2. **Expansion to More Economic Indicators**
   - Include additional variables such as Human Development Index (HDI), corruption index, income inequality, or CO_2 emissions for multi-dimensional analysis.
3. **Machine Learning Integration**

- Predict economic trends using historical data and train models to forecast future index scores.

4. **User Personalization and Reporting**
    - Enable users to create custom dashboards, save views, and generate automated country-wise reports.
5. **Mobile-Friendly Dashboards**
    - Optimize dashboards for smartphones and tablets to make economic insights accessible on the go.
6. **Collaboration and Feedback Features**
    - Allow multiple users (e.g., teams or institutions) to collaborate, comment, or provide feedback directly on the dashboard.
7. **Multilingual Support**
    - Translate the platform into multiple languages to reach global audiences, especially in educational institutions.

11. APPENDIX

DatasetLink:
https://drive.google.com/file/d/1EBIa1LtM3Ni2Uh3nekLB6wt3263Q3NeX/view?usp=share_link

GitHub & Project Demo Link: https://github.com/LeelaKumari-LK/Measuring-the-Pulse-of-Prosperity-An-Index-of-Economic-Freedom-Analysis